Smart Clinic AI Assistant

Project Proposal

Initiative: DEPI — Digital Egypt Pioneers Initiative **Track:** AI & Data Science **Team Leader:** Ahmed Khaled Abdelfattah

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1. Project Overview

Smart Clinic AI Assistant is an n8n-based automation system that manages a clinic's communication with patients and streamlines internal operations. The system integrates a Telegram chatbot, an LLM for natural-language understanding, Google Calendar for scheduling, and Google Sheets as the data store, with the ability to handle more than one doctor in the clinic. It reduces manual work for clinic staff, improves the patient experience with 24/7 availability, and gives the doctor(s) an accurate, automated view of daily clinic performance.

2. Project Objectives

1. **Automated Patient Inquiries** — Answer recurring patient questions (location, fees, working hours) via Telegram, around the clock.
2. **Appointment Booking** — Let patients self-book appointments with real-time availability checks against the calendar, accounting for each doctor's individual schedule and specialization.
3. **Automated Reminders** — Send an interactive reminder 24 hours before each appointment (Confirm / Reschedule / Cancel) to reduce no-shows.
4. **Financial Reporting** — Generate an automated daily report for the doctor(s) with total revenue and case count, calculated programmatically (not by the LLM) to guarantee accuracy.
5. **Patient Satisfaction Analysis** — Collect post-visit feedback via Telegram, classify it (Positive / Neutral / Negative), and surface it to support service improvement.

3. Project Scope

In scope for the MVP:

- A Telegram-based conversational interface for patients.
- Intent classification between general inquiries and booking requests.
- Appointment booking and availability checks against Google Calendar, supporting multiple doctors.
- 24-hour interactive appointment reminders with status tracking.
- A daily, code-calculated financial summary emailed to the doctor(s).
- Post-visit feedback collection and sentiment classification.
- Data storage via Google Sheets (patients, appointments, daily ledger, feedback).

Out of scope for the MVP (candidate future work):

- Migration to a relational database (e.g., PostgreSQL) if data volume or concurrency needs grow.
- WhatsApp integration (Telegram is used for the MVP due to lower setup overhead).
- Handling of non-text patient messages (voice notes, images) beyond a basic fallback reply.
- Multi-branch / multi-clinic support.

4. Target Users

- **Patients** — interact with the system via Telegram to ask questions, book appointments, respond to reminders, and give feedback.
- **Doctor(s)** — receive the daily financial report and benefit from fewer no-shows and better visibility into clinic performance.
- **Clinic secretary/staff** — continue to record payments and attendance in the shared ledger that the system reads from.

5. Proposed Solution

The system is built as a set of n8n workflows orchestrating a Telegram bot, an LLM, Google Calendar, Google Sheets, and Gmail:

Module	Trigger	Summary
1. Message Handling & Booking	Telegram message	Classifies intent (inquiry vs. booking); answers inquiries from clinic info; collects patient data and books appointments against calendar availability per doctor
2. Reminders & Follow-up	Daily/periodic schedule	Sends a 24h-ahead interactive reminder; updates booking status based on patient response (Confirmed / Cancelled / Needs Rescheduling)
3. Daily Financial Report	End-of-day schedule	Reads the day's ledger, computes totals via code (not LLM), and emails a summary to the doctor(s)
4. Patient Feedback & Sentiment	Scheduled after appointment	Requests feedback via Telegram, classifies sentiment, and logs it for review

Data store: Google Sheets is used for the MVP (Bookings, Daily Ledger, Feedback) for fast setup and team accessibility, with a logical data model designed so it can be migrated to a relational database later without a redesign.

6. Assumptions & Open Items

- Clinic details (number of doctors, working hours, appointment duration, cancellation policy) will be confirmed with the client/instructor before final design.
- Deployment target (working prototype vs. production-ready system) to be confirmed.
- Telegram is assumed as the patient-facing channel for the MVP; WhatsApp is a possible future direction.

Project Plan

Timeline (Milestones)

Aligned to the official DEPI submission dates for this track.

Milestone	Deliverables Due	Date
M1 - Planning & Management	Project Proposal, Project Plan, Task Assignment, Risk Assessment, KPIs	16 Oct 2026
M2 - Requirements & Literature Review	Stakeholder Analysis, User Stories, Functional/Non-functional Requirements, Literature Review	16 Oct 2026
M3 - System Analysis & Design	Use Case/Sequence/DFD diagrams, ER Diagram & Schema, Wireframes, Architecture	6 Nov 2026
M4 - Implementation	Working n8n workflows, source code/export on GitHub, README	30 Nov 2026
M5 - Final Presentation & Reports	User Manual, Technical Documentation, Presentation, Infographic	4 Dec 2026

Gantt Overview (by week)

Phase	Weeks	Focus
Requirements & Design	Weeks 1-3 (to 6 Nov)	Confirm clinic requirements, finalize scope, produce all System Analysis & Design artifacts
Core Build	Weeks 4-6	Module 1 (Telegram Assistant) and Module 2 (Booking)
Extended Build	Weeks 6-7	Module 3 (Reminders) and Module 4 (Financial Report)
Final Build & QA	Week 7 (to 30 Nov)	Module 5 (Feedback), end-to-end testing, bug fixes
Wrap-up	Week 8 (to 4 Dec)	Documentation, presentation, infographic

Resource Allocation

- Team Leader (self): planning, architecture decisions, documentation review, integration oversight.
- Member 2: Telegram bot + LLM intent/response logic.
- Member 3: n8n workflow automation (Calendar, Sheets, Gmail integrations).
- Members 4 & 5: testing, bug tracking, documentation support, presentation.

Pending confirmation of each member's actual skills and availability - see Task Assignment below.

Task Assignment & Roles

Member	Primary Responsibility	Key Deliverables
Ahmed Khaled Abdelfattah (Team Lead)	Planning, architecture, documentation, integration review	All Planning & Management docs, final architecture sign-off
Ahmed Hossam Abdullah	n8n automation & integrations (Modules 2-3)	Router/classifier, inquiry & sentiment prompts
Mohamed Alaa Ibrahim	Telegram bot & LLM logic (Modules 1, 4-5)	Calendar/Sheets booking and reminder workflows
Ziad Ahmed Fathy	Testing & QA	Test cases, bug reports
Ahmed Mossa Anwar	Documentation & presentation support	User manual, slides, infographic

This split is a starting proposal and will be adjusted once each member confirms their technical background and available hours.

Risk Assessment & Mitigation Plan

Risk	Likelihood	Impact	Mitigation
Low team member engagement	Medium	High	Confirm availability early; redistribute tasks among active members if needed

Risk	Likelihood	Impact	Mitigation
Delays building n8n workflows	Medium	Medium	Start with a minimal working version per module before adding features
Integration issues between services	Medium	Medium	Test each integration in isolation before connecting modules
Double-booked appointments	Low	High	Availability check always goes through Calendar/Code Node, never the LLM alone
Financial report errors	Low	High	All calculations done in a Code Node, not by the LLM
Inaccurate AI responses	Medium	Medium	Keep the LLM's role limited to language understanding/generation; business logic stays in code
Patient data protection	Medium	High	Limit data collected to what's needed; restrict booking actions to the requesting patient's own chat_id
Missing the final deadline	Low	High	Follow the milestone schedule above; flag slippage early to the team

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KPIs (Key Performance Indicators)

KPI	Target	How It's Measured
Response Time	Reply within a few seconds of a patient message	Timestamp difference, Telegram message to bot reply
Inquiry Resolution Rate	High share of inquiries answered without human handoff	Ratio of inquiries answered by Module 1 vs. escalated
Booking Success Rate	High share of booking attempts completed successfully	Confirmed bookings / total booking attempts
Reminder Delivery Rate	Reminders sent for all confirmed appointments	Reminders sent / appointments due

KPI	Target	How It's Measured
Financial Report Accuracy	Report totals match the ledger exactly	Automated total vs. manual spot-check
Feedback Collection Rate	Reasonable share of patients respond to the feedback request	Feedback responses / feedback requests sent
Milestone Completion Rate	All milestones delivered on or near their due date	Milestones completed on time / total milestones

Exact target percentages to be set once the MVP is running and we have baseline numbers from testing.